

Project proposal - Group 2

Background scenario of the organisation:

A higher educational institution (e.g. a university).

Project goal:

To identify struggling students, who are more likely to dropout so that timely intervention can be made by institutions administrators and faculty while maintaining ethical and transparent data practices. This will be done by implementing a Machine Learning model to predict which students are at risk of dropping out based on demographic, academic and socio-economic indicators.

Measures and actions:

These students would be contacted and supported based on individual requirements. Therapeutic, Tutoring, Financial, etc. aid could be offered.

Data used for the project:

[Predict Student's Dropout and Academic Success dataset](#) - Valentim Realinho, Mónica Vieira Martins, Jorge Machado, Luís Baptista

Dataset created from a higher education institution (acquired from several disjoint databases) related to students enrolled in different undergraduate degrees, such as agronomy, design, education, nursing, journalism, management, social service, and technologies.

The dataset includes information known at the time of student enrollment (academic path, demographics, and social-economic factors) and the students' academic performance at the end of the first and second semesters

Stakeholders and impact:

Students may hesitate to reach out to the faculty, or may otherwise be in difficult situations in which they need help but do not receive it on time. There can be multitudes of reasons why students might drop out of school, leading to a cascading chain of negative effects towards decrease in future quality of life and on the economy overall. The project could help with the identification of students at risk. It also helps the institution maintain stable academic performance.

Benefits of the project:

Hope of this project is to help with decrease in student dropout rate by early identification and hence timely intervention.

Problems and concerns:

What problems or concerns could arise in connection with this project

The project naturally requires data on student performance and background. This creates concerns about privacy, we need to make sure no individually identifiable information is used. This is difficult as multiple characteristics can be combined, like the grades and evaluation in conjunction with other features to identify individual students.

Another potential concern is bias, we need to make sure that the model does not bias evaluations based on protected characteristics, like nationality, gender etc.

Issues could also arise from underrepresented groups in the dataset. Our model might have a worse performance detecting students at risk of dropout from minority groups in the dataset.